

Document database

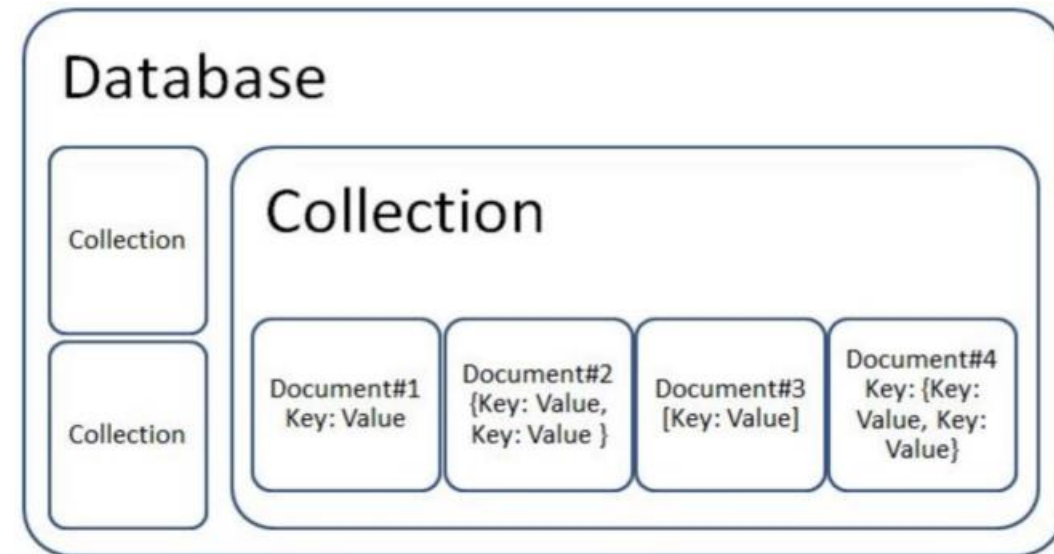
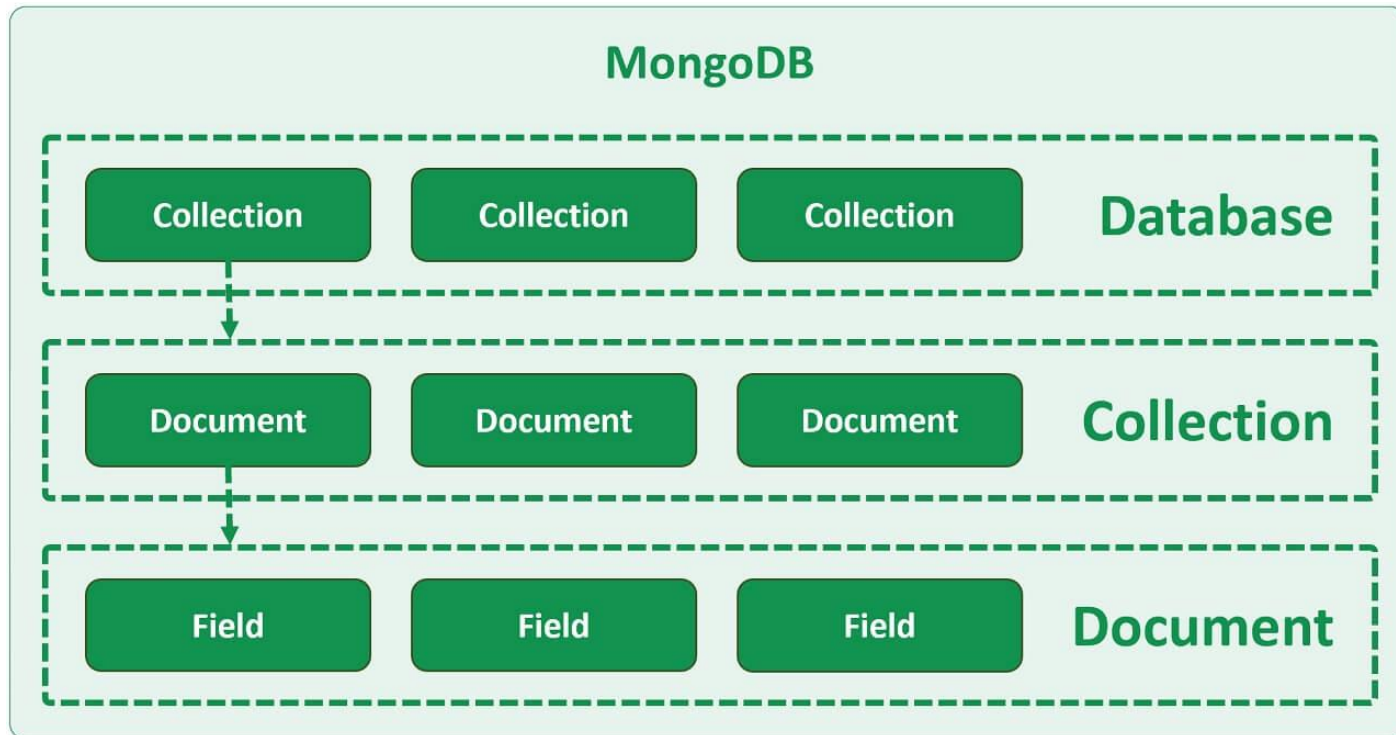
MongoDB

Mongo DB คืออะไร



- ฐานข้อมูลเชิงเอกสาร (Document store)
- เก็บข้อมูลขนาดใหญ่ที่มีความยืดหยุ่น
- ง่ายต่อการปรับทำงานข้าม platform
- จัดเก็บข้อมูลในรูปแบบของ JSON แล้วเซฟไว้ในเอกสารรูปแบบไบนารี BSON

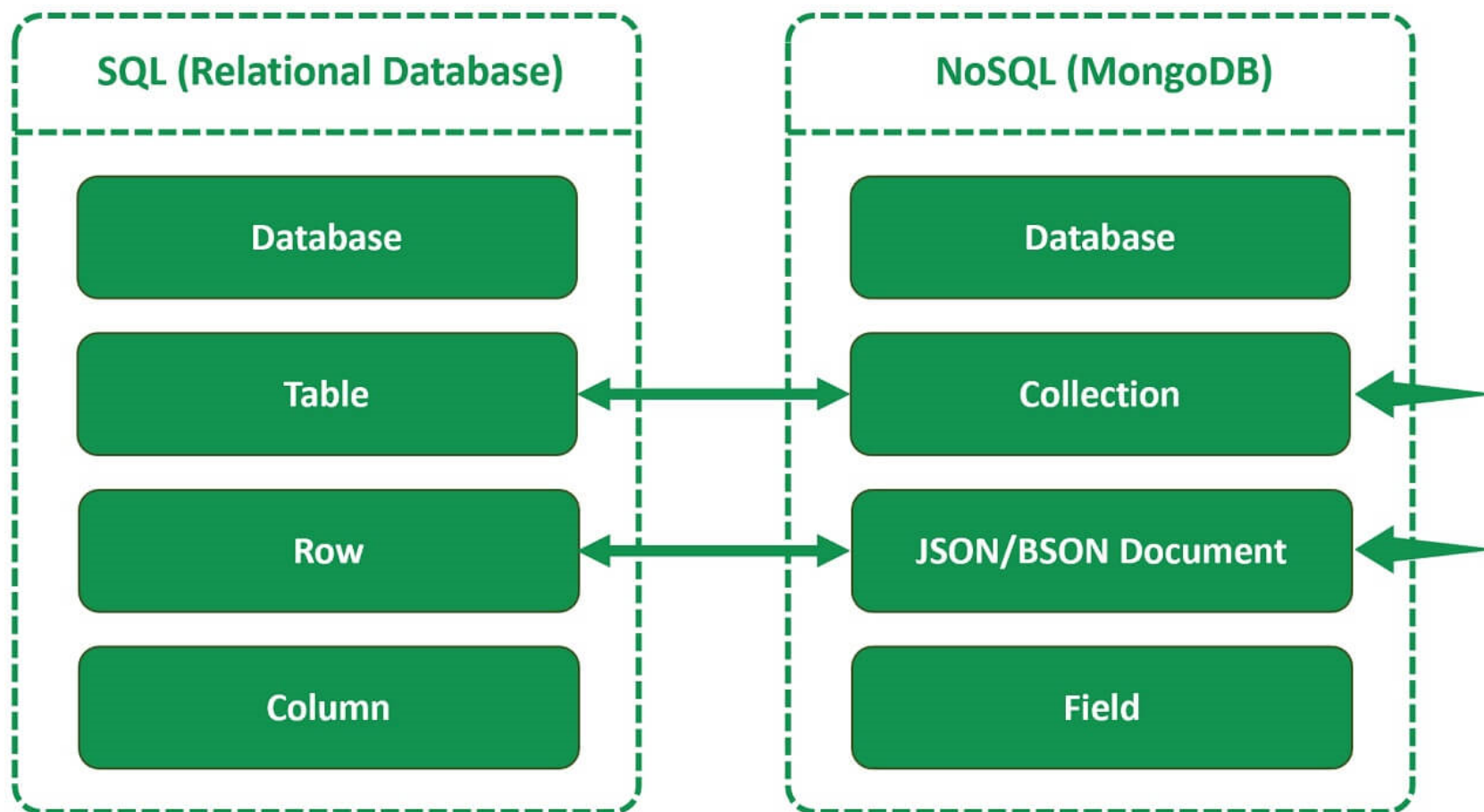
โครงสร้างการจัดเก็บข้อมูล



โครงสร้างการจัดเก็บข้อมูลของ MongoDB

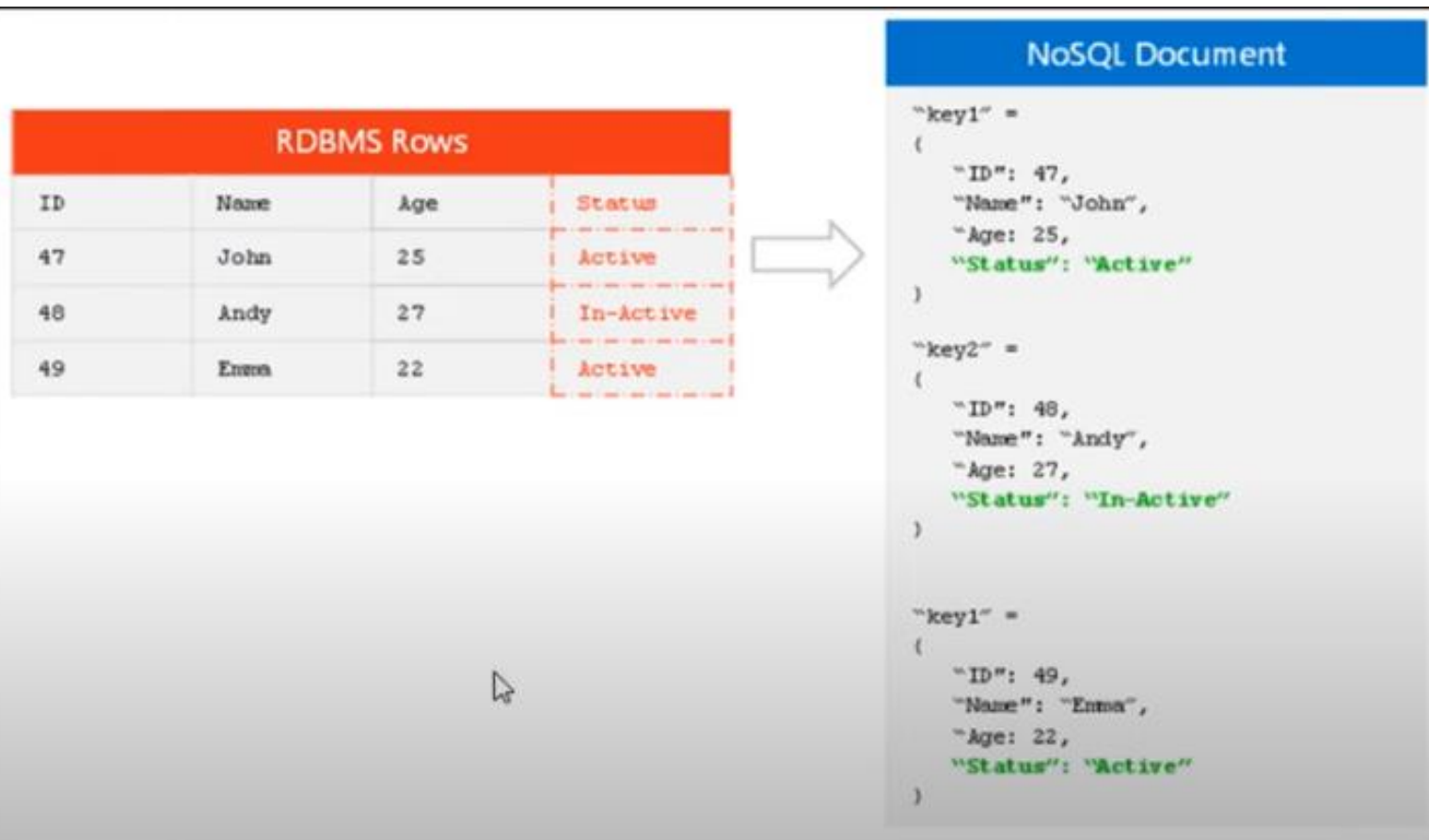
- Database (ฐานข้อมูล) เป็นส่วนที่ใช้เก็บชุด Collection หรือชุดข้อมูล
- Collection หรือชุดข้อมูลเปรียบเทียบกับตารางใน RDB
- Document หรือเอกสารที่จัดเก็บข้อมูลเป็นคู่ของ key และ value

เปรียบเทียบโครงสร้าง RDB และ MongoDB



Primary key

- RDB
 - PK เป็นค่าที่ระบุการจัดเก็บข้อมูลในแต่ละแถว และต้องไม่ซ้ำกัน
- MongoDB
 - ObjectId เป็นค่าที่ไม่ซ้ำกัน ถ้าไม่ระบุ ฐานข้อมูลจะสร้างให้อัตโนมัติ



JSON Document

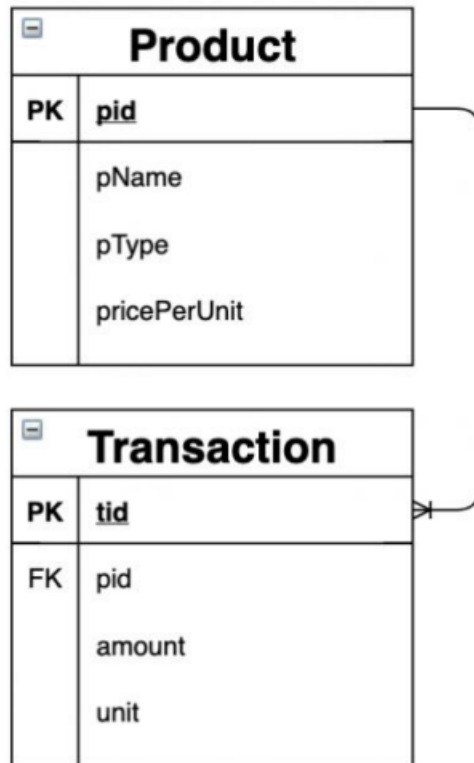
```
{
  "_id": 1,
  "name": "John",
  "age": 25,
  "gender": "Male",
  "graduated": true,
  "address": {
    "street": "123 Main Street",
    "city": "Star City"
  },
  "awards": [ {"award": "Star Student", "year": 2021}, {"award": "Math Master", "year": 2020} ]
}
```



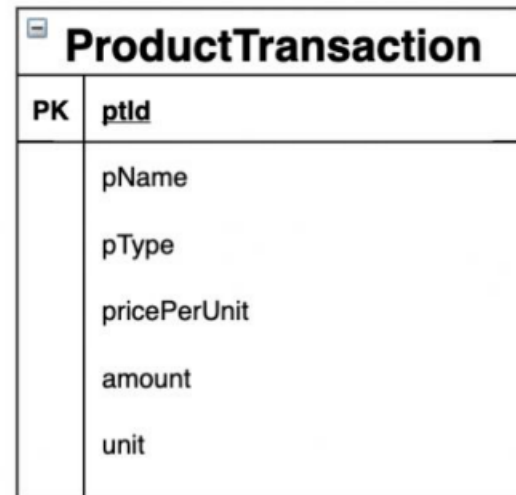
Nested JSON document

Document DB

Normalization



Denormalization

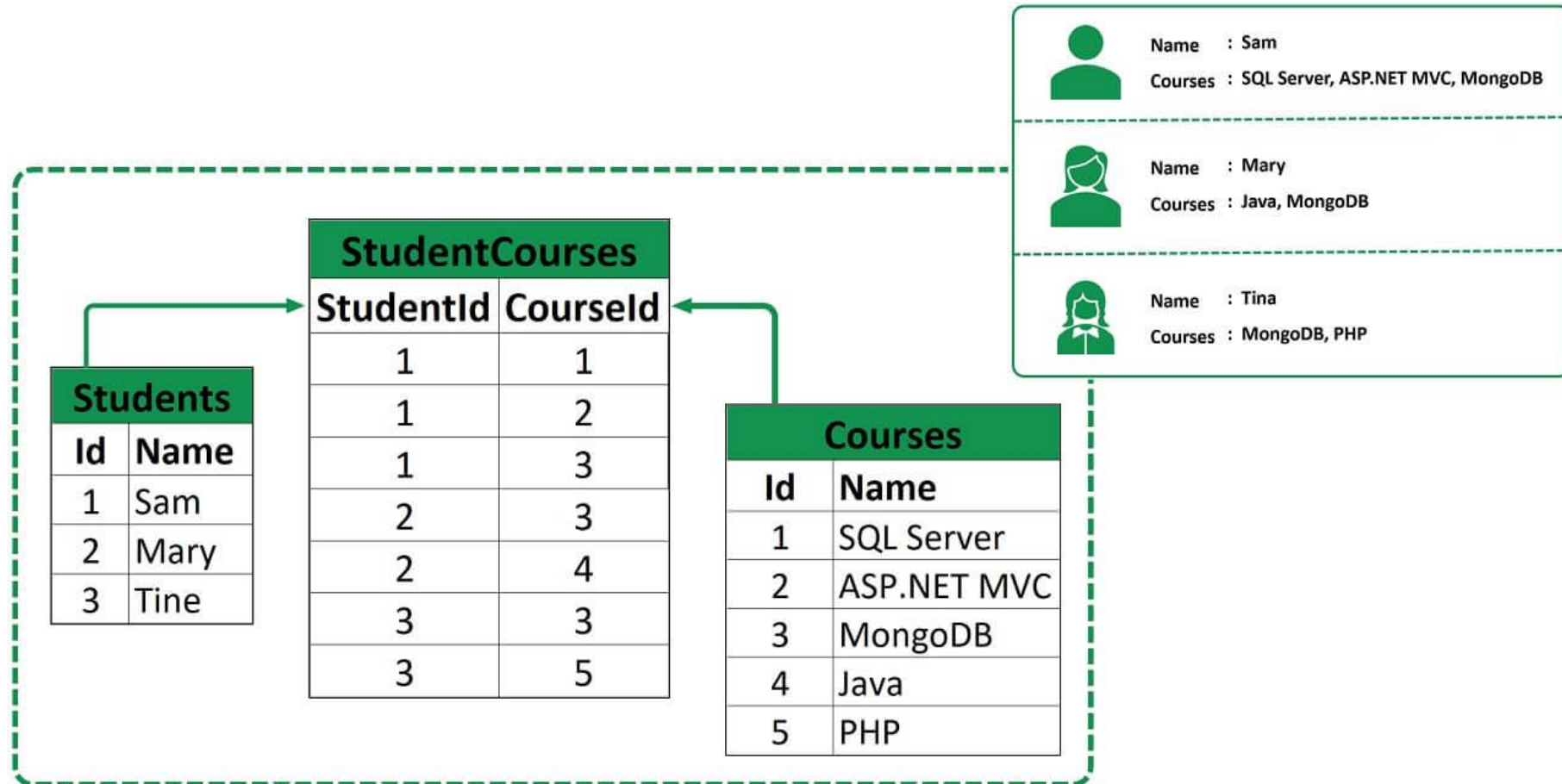


Simple use case

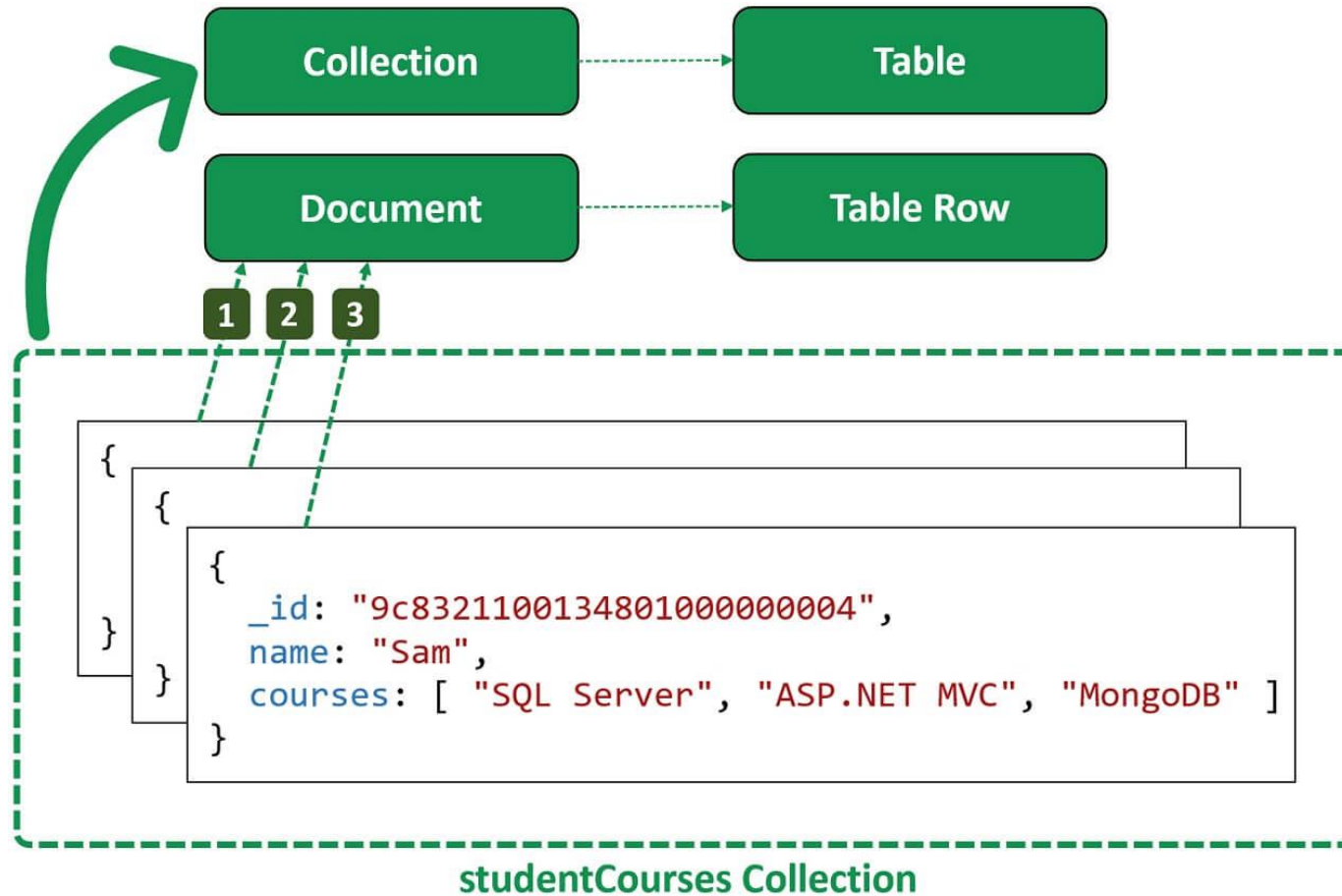
- Develop an application for displaying student names and the courses that they have enrolled in, on a web page

	Name : Sam Courses : SQL Server, ASP.NET MVC, MongoDB
	Name : Mary Courses : Java, MongoDB
	Name : Tina Courses : MongoDB, PHP

Relational database



Document DB



Relationships

- Relationships between document
- Reference between document

Relationship

```
{
  title: "MongoDB: The Definitive Guide",
  author: [ "Kristina Chodorow", "Mike Dirolf" ],
  published_date: ISODate("2010-09-24"),
  pages: 216,
  language: "English",
  publisher: {
    name: "O'Reilly Media",
    founded: 1980,
    location: "CA"
  }
}

{
  title: "50 Tips and Tricks for MongoDB Developer",
  author: "Kristina Chodorow",
  published_date: ISODate("2011-05-06"),
  pages: 68,
  language: "English",
  publisher: {
    name: "O'Reilly Media",
    founded: 1980,
    location: "CA"
  }
}
```

Reference

```
{
  name: "O'Reilly Media",
  founded: 1980,
  location: "CA",
  books: [123456789, 234567890, ...]
}

{
  _id: 123456789,
  title: "MongoDB: The Definitive Guide",
  author: [ "Kristina Chodorow", "Mike Dirolf" ],
  published_date: ISODate("2010-09-24"),
  pages: 216,
  language: "English"
}
```

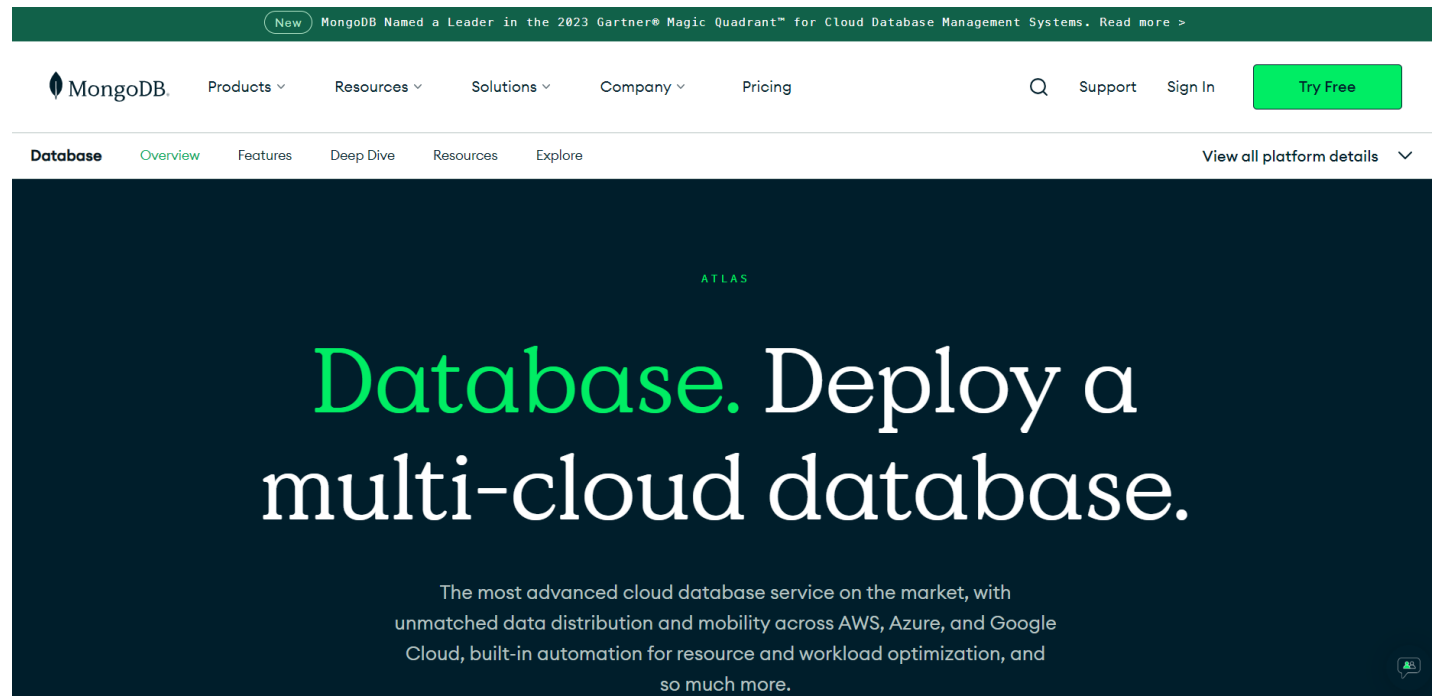
Installation MongoDB

- Server
- Cloud



Get Started with MongoDB Atlas

- **Registration** : <https://www.mongodb.com/atlas/database>
- Click: SignIn

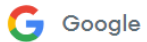


Signup with gmail



Log in to your account

Don't have an account? [Sign Up](#)



Or with email and password

Email Address

Next

MongoDB.local is coming to a city near you!

Enjoy technical deep-dives, one-on-one expert advice, and product tutorials to elevate your skills.

Enjoy a 50% discount with code WEB50.

[View the schedule](#) →



Accept term of service



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Please acknowledge the following terms and conditions to finish creating your account.

☒ I accept the [Privacy Policy](#) and the [Terms of Service](#)

Cancel Signup

Submit

Overview

 **DEPLOYMENT**

Database

Data Lake

 **SERVICES**

Device & Edge Sync

Triggers

Data API

Data Federation

Atlas Search

Stream Processing

Migration

 **SECURITY**

Backup

Database Access

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Clusters



Create a cluster

Choose your cloud provider, region, and specs.

Build a Cluster

Once your database is up and running, live migrate an existing MongoDB database into Atlas with our [Live Migration Service](#).

Create a cluster

Deploy your cluster

Use a template below or set up advanced configuration options. You can also edit these configuration options once the cluster is created.

☐ **M10** **\$0.09/hour**
For production applications with sophisticated workload requirements.

STORAGE	RAM	vCPU
10 GB	2 GB	2 vCPUs

☐ **Serverless** **\$0.12/1M reads**
For application development and testing, or workloads with variable traffic.

STORAGE	RAM	vCPU
Up to 1TB	Auto-scale	Auto-scale

☒ **M0** **Free**
For learning and exploring MongoDB in a cloud environment.

STORAGE	RAM	vCPU
512 MB	Shared	Shared

✓ **Free forever!** Your M0 cluster is ideal for experimenting in a limited sandbox. You can upgrade to a production cluster anytime.

Name

You cannot change the name once the cluster is created.

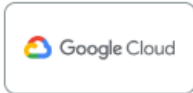
☐ Automate security setup ⓘ

☒ Preload sample dataset ⓘ

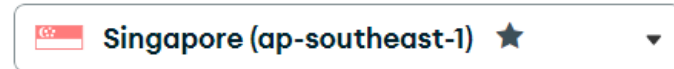
Deploy บนเครื่องแบบไหน

Create a cluster

Provider



Region



★ Recommended ⓘ 🌿 Low carbon emissions ⓘ

Tag (optional)

Create your first tag to categorize and label your resources; more tags can be added later. [Learn more.](#)

:

I'll do this later

Go to Advanced Configuration

Create Deployment

Security

Only an IP address you add to your Access List will be able to connect to your project's clusters.

You can edit, delete, or add IP addresses in [Network Access](#).



Add my current IP Address



Allow access from anywhere



Manually enter IP Address or CIDR notation

Add Connection



Create a database user

This first user will have [atlasAdmin](#) permissions for this project. You'll need your database user's credentials to connect to your cluster.

We autogenerated an username and password. You can use this or create your own. You can edit, delete, or add database users in [Database Access](#).

Username

342db

Password

.....



Copy

Create User

Connection

Connect to 432Cluster

Configure your connection below and follow the instructions on the left.
Instructions will be updated based on your configurations.

i **You can't connect yet.** Set up your firewall access and user security permissions first. Then you'll be able to see your connection options.

i Add a connection IP address

Only an IP address you add to your Access List will be able to connect to your project's clusters.

You can edit, delete, or add IP addresses in [Network Access](#) .

☒ Add my current IP Address

☐ Allow access from anywhere

☐ Manually enter IP Address or CIDR notation

Add Connection

Database Access

Data Lake

 SERVICES

Device & Edge Sync

Triggers

Data API

Data Federation

Atlas Search

Stream Processing

Migration

 SECURITY

Quickstart

Backup

Database Access

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Database Access

Database Users

Custom Roles

User Name ↕	Authentication Method ▲	MongoDB Roles
 342db	SCRAM	readWriteAnyDatabase@admin

Network access

SERVICES

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Advanced

Network Access

IP Access ListPeeringPrivate Endpoint

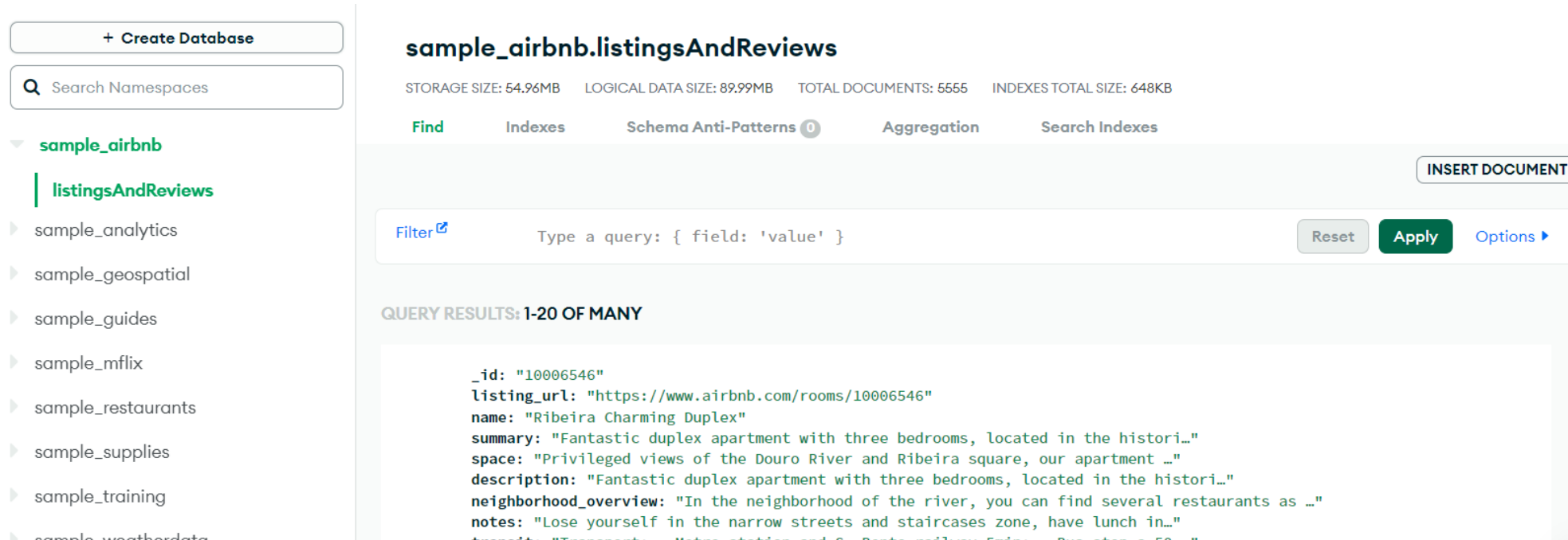
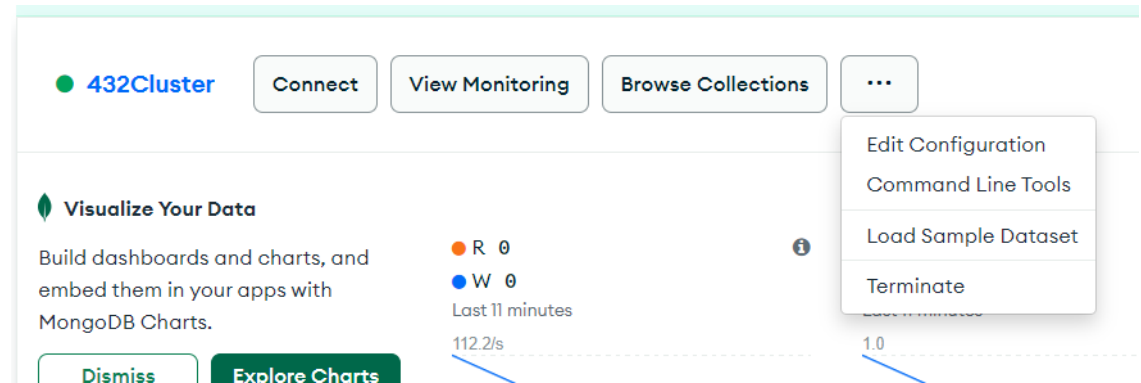
You will only be able to connect to your cluster from the following list of IP Addresses:

IP Address	Comment	Status
184.22.141.199/32 (includes your current IP address)		● Active

System Status: All Good

Load Sample data

- Click ... then load sample data
- Click brose connection to open



Sample_nfix

+ Create Database

Q Search Namespaces

▶ sample_airbnb

▶ sample_analytics

▶ sample_geospatial

▶ sample_guides

▼ sample_mflix

comments

embedded_movies

movies

sessions

sample_mflix

LOGICAL DATA SIZE: 115.94MB STORAGE SIZE: 92.84MB INDEX SIZE: 18.6MB TOTAL COLLECTIONS: 6

CREATE COLLECTION

Collection Name	Documents	Logical Data Size	Avg Document Size	Storage Size	Indexes	Index Size	Avg Index Size
comments	41079	11.14MB	285B	6.13MB	1	1.16MB	1.16MB
embedded_movies	3483	71.9MB	21.14KB	67.72MB	1	148KB	148KB
movies	21349	32.54MB	1.56KB	18.77MB	2	17.1MB	8.55MB
sessions	1	540B	540B	20KB	2	40KB	20KB
theaters	1564	341.63KB	224B	168KB	2	112KB	56KB
users	185	28.88KB	160B	36KB	2	48KB	24KB

<https://www.mongodb.com/docs/atlas/sample-data/sample-mflix/>

Query your data using filter tool

[Filter](#) 

Type a query: { field: 'value' }

[Reset](#) [Apply](#) [Options ▶](#)

Match by a Single Condition

- Query filter finds all documents where title is Jurassic Park

Filter 

```
{ "title": "Jurassic Park" }
```

QUERY RESULTS: 1-1 OF 1

```
_id: ObjectId('573a1399f29313caabcedc5d')
fullplot: "Huge advancements in scientific technology have enabled a mogul to cre..."
▸ imdb: Object
  year: 1993
  plot: "During a preview tour, a theme park suffers a major power breakdown th..."
▸ genres: Array (3)
  rated: "PG-13"
  metacritic: 68
  title: "Jurassic Park"
  lastupdated: "2015-08-31 00:04:50.280000000"
▸ languages: Array (2)
▸ writers: Array (3)
  type: "movie"
▸ tomatoes: Object
  poster: "https://m.media-amazon.com/images/M/MV5BMjM2MDgxMDg0Nl5BMl5BanBnXkFtZT..."
```

- Query filter finds all documents where runtime is 95

Filter [🔗](#)

```
{"runtime":95}
```

QUERY RESULTS: 1-20 OF MANY

```
_id: ObjectId('573a1391f29313caabcd7b98')
plot: "A bitter clown endeavors to rescue the young woman he loves from the l..."
▸ genres: Array (3)
  runtime: 95
▸ cast: Array (4)
  num_mflix_comments: 0
poster: "https://m.media-amazon.com/images/M/MV5B0TEwMTA10Tc1NF5BMl5BanBnXkFtZT..."
title: "He Who Gets Slapped"
fullplot: "Story of an inventor who, suffering betrayal in life, makes a career o..."
▸ languages: Array (1)
```

- Query filter finds all documents where year is less than or equal 2000

Filter [🔗](#)

```
{"year" : {$lte : 1921}}
```

... respect when three shaggy men are responsible for the death of the father and

- ▶ countries: Array (1)
- released: 1921-12-31T00:00:00.000+00:00
- ▶ directors: Array (1)
- ▶ writers: Array (3)
- ▶ awards: Object
- lastupdated: "2015-08-23 01:12:08.943000000"
- year: 1921
- ▶ imdb: Object
- type: "movie"
- ▶ tomatoes: Object
- num_mflix_comments: 0

Query operators

{ field: { \$operator: value } }

{ age: { \$lte: 30 } }

\$eq	Equal
\$ne	Not equal
\$gt	Greater than
\$gte	Greater than or equal to
\$lt	Less than
\$lte	Less than or equal to
\$in	In
\$nin	Not in

- Find all document year in [1921, 1917]

Filter [🔗](#)

```
{"year": {$in: [1921,1917]}}
```

```
plot: "A young man, unaccustomed to children, must accompany a young girl on ..."  
▶ genres: Array (2)  
runtime: 35  
rated: "PASSED"  
▶ cast: Array (3)  
poster: "https://m.media-amazon.com/images/M/MV5BYjgzYzY1NjEtYWQxZS00ZjA4LWJlYm..."  
title: "Now or Never"  
fullplot: "Mary is looking after a young child whose parents have little time for..."  
▶ languages: Array (1)  
released: 1921-03-27T00:00:00.000+00:00  
▶ directors: Array (2)  
▶ writers: Array (2)  
▶ awards: Object  
lastupdated: "2015-06-25 00:02:16.613000000"  
year: 1921  
... " " ...
```

Find projection

- Queries in MongoDB return all fields in matching documents.
- projection is used to limit the amount of data.
- A projection basically tells what fields to include or exclude in the matching documents.
- Use 1 to include a field and 0 to exclude a field.

Projection

Filter 🔗	<code>{"year": {\$lte: 1917}}</code>
Project	<code>{"title" : 1}</code>
Sort	<code>{ field: -1 } or [['field', -1]]</code>
Collation	<code>{ locale: 'simple' }</code>

```
_id: ObjectId('573a1390f29313caabcd6377')  
title: "Wild and Woolly"
```

```
_id: ObjectId('573a1390f29313caabcd56df')  
title: "The Italian"
```

```
_id: ObjectId('573a1390f29313caabcd4803')
```

Filter 🔗	<code>{"year": {\$lte: 1917}}</code>
Project	<code>{"title" : 1, "runtime" : 1}</code>
Sort	<code>{ field: -1 } or [['field', -1]]</code>
Collation	<code>{ locale: 'simple' }</code>

QUERY RESULTS: 1-16 OF 16

```
_id: ObjectId('573a1390f29313caabcd6377')  
runtime: 72  
title: "Wild and Woolly"
```

```
_id: ObjectId('573a1390f29313caabcd56df')  
runtime: 78  
title: "The Italian"
```

Nested object

- Awards is a nested JSON document

```
▸ directors: Array (1)
▼ writers: Array (4)
  0: "Leonid Andreyev (adapted from the play by)"
  1: "Carey Wilson (adapted for the screen by)"
  2: "Victor Sjöström (adapted for the screen by)"
  3: "Marian Ainslee (titles)"
▼ awards: Object
  wins: 1
  nominations: 0
  text: "1 win."
  lastupdated: "2015-09-11 00:00:29.303000000"
  year: 1924
▸ imdb: Object
▸ countries: Array (1)
```

- Find all document where wins is 1

Filter 

`{"awards.wins" : 1}`

```

title: "He Who Gets Slapped"
fullplot: "Story of an inventor who, suffering betrayal in life, makes a career
▸ languages: Array (1)
  released: 1924-12-22T00:00:00.000+00:00
▸ directors: Array (1)
▸ writers: Array (4)
▾ awards: Object
  wins: 1
  nominations: 0
  text: "1 win."
lastupdated: "2015-09-11 00:00:29.303000000"
year: 1924
▸ imdb: Object
▸ countries: Array (1)

```

Filter 

`{"awards.wins" : 1}`

Project `{"title":1, "awards" :1}`

Sort `{ field: -1 } or [['field', -1]]`

Collation `{ locale: 'simple' }`

```

title: "He Who Gets Slapped"
▸ awards: Object

```

```

_id: ObjectId('573a1390f29313caabcd6377')
title: "Wild and Woolly"
▸ awards: Object

```

Sort

- Use -1 in descending order
- Use 1 in ascending order

```
Filter 🔗      {"awards.wins" : 1}
Project      {"title":1, "awards" :1}
Sort         {"title" :1}
Collation    { locale: 'simple' }
```

```
title: "'Doc'"
▸ awards: Object
```

```
_id: ObjectId('573a1393f29313caabcdc5dc')
title: "'Pimpernel' Smith"
```

Multiple condition

- { \$and: [{ scores: 75, name: "Greg Powell" }] }
- { \$or: [{ version: 4 }, { name: "Andrea Le" }] }
- { name: { \$not: { \$eq: "Andrea Le" } } }

ปฏิบัติการครั้งที่ 1